1.4 Modeling with Linear Functions

Name _____ Period ____ Date _____

Algebra II 1.4 (30-minute plan) · Student sheet

I can…

- write equations of linear functions from graphs and tables.
- find a line of fit and the line of best fit.
- interpret the correlation coefficient and use linear models.

Vocabulary · fill in the missing word

Slope-intercept form — $y =$ _____

Point-slope form — $y - y_1 =$ _____

Line of fit — a line drawn close to most of the points of a _____ plot

Line of best fit — the line that fits the data most _____ (from technology)

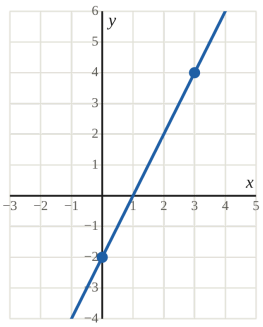
Correlation coefficient r — a number from −1 to 1 that _____ how well a line fits; near ±1 is strong, near 0 is weak

Part 1 · Notes and examples (fill in each note, then work the example)

Equations of lines: Slope m = _____; $y = mx + b$; $y - y_1 =$ _____

Example 1

Write an equation of the line.



Line of fit: A line of fit is drawn close to _____ of the points.

Example 2

Use technology to find the line of best fit for the study-time data and the correlation coefficient. Predict the score for 10 hours of studying.

Hours studied	1	2	3	4	5	6	7	8
Score	62	68	71	75	80	83	88	90

Correlation coefficient: r near 1: _____. r near −1: _____. r near 0: _____.

Example 3

Describe the correlation shown by each correlation coefficient.

a. $r = -0.94$ b. $r = 0.12$ c. $r = 0.85$

Part 2 · Partner practice

Partner practice

Write an equation of the line through $(-1, 7)$ and $(3, -1)$.